HW 04 - Predict coffee preferences

HOMEWORK

MODIFIED
October 3, 2025

> **Important**
> This homework is due October 8 at 11:59pm ET.

Learning objectives

- Implement data cleaning and preparation
- Define data preprocessing and feature engineering steps for predictive models
- Estimate a range of model types
- Creatively explore alternative model workflows
- Utilize tuning parameters to increase model performance
- Evaluate model performance

Getting started

- Go to the **info4940-fa25** organization on GitHub. Click on the repo with the prefix **hw-04**. It contains the starter documents you need to complete the lab.

- Clone the repo and start a new workspace in Positron. See the Homework 0 instructions for details on cloning a repo and starting a new R project.

General guidance

> **Guidelines + tips**
>
> - Set your random seed to ensure reproducible results.
> - Use caching to speed up the rendering process.
> - Use **parallel processing** to speed up rendering time. Note that this works differently on different systems and operating systems, and it also makes it harder to debug code and track progress in model fitting. Use at your own discretion.
>
> Remember that continuing to develop a sound workflow for reproducible data analysis is important as you complete this homework and other assignments in this course. There will be periodic reminders in this assignment to remind you to render, commit, and push your changes to GitHub. You should have **at least 3 commits with meaningful commit messages** by the end of the assignment.

> **Workflow + formatting**
>
> Make sure to
>
> - Update author name on your document.
> - Label all code chunks informatively and concisely.
> - Follow consistent code style guidelines.
> - Make at least 3 commits.
> - Resize figures where needed, avoid tiny or huge plots.
> - Turn in an organized, well formatted document.

> **Presenting results of multiple models**
>
> For the love of all that is pure in this world, please consider how to present the results of your modeling efforts. Do not just rely on raw output from R to tell us what we need to know.
>
> - Your plots should include an informative title, axes should be labeled, and careful consideration should be given to aesthetic choices.
> - Maybe condense information into one or a handful of custom graphs.
> - You can create simple formatted tables using {gt}/ `great_tables`

The Great American Coffee Tasting

In October 2023, James Hoffmann and coffee company Cometeer held the "Great American Coffee Taste Test" on YouTube, during which viewers were asked to fill out a survey about 4 coffees they ordered from Cometeer for the tasting. Tidy Tuesday published the data set we are using.

It includes the following features:

variable	class	description
submission_id	character	Submission ID
age	character	What is your age?
cups	character	How many cups of coffee do you typically drink per day?
where_drink	character	Where do you typically drink coffee?
brew	character	How do you brew coffee at home?
brew_other	character	How else do you brew coffee at home?
purchase	character	On the go, where do you typically purchase coffee?
purchase_other	character	Where else do you purchase coffee?
favorite	character	What is your favorite coffee drink?
favorite_specify	character	Please specify what your favorite coffee drink is
additions	character	Do you usually add anything to your coffee?
additions_other	character	What else do you add to your coffee?
dairy	character	What kind of dairy do you add?
sweetener	character	What kind of sugar or sweetener do you add?
style	character	Before today's tasting, which of the following best described what kind of coffee you like?
strength	character	How strong do you like your coffee?
roast_level	character	What roast level of coffee do you prefer?
caffeine	character	How much caffeine do you like in your coffee?
expertise	numeric	Lastly, how would you rate your own coffee expertise?
coffee_a_bitterness	numeric	Coffee A - Bitterness
coffee_a_acidity	numeric	Coffee A - Acidity
coffee_a_personal_preference	numeric	Coffee A - Personal Preference
coffee_a_notes	character	Coffee A - Notes
coffee_b_bitterness	numeric	Coffee B - Bitterness
coffee_b_acidity	numeric	Coffee B - Acidity
coffee_b_personal_preference	numeric	Coffee B - Personal Preference
coffee_b_notes	character	Coffee B - Notes
coffee_c_bitterness	numeric	Coffee C - Bitterness
coffee_c_acidity	numeric	Coffee C - Acidity
coffee_c_personal_preference	numeric	Coffee C - Personal Preference
coffee_c_notes	character	Coffee C - Notes
coffee_d_bitterness	numeric	Coffee D - Bitterness
coffee_d_acidity	numeric	Coffee D - Acidity
coffee_d_personal_preference	numeric	Coffee D - Personal Preference
coffee_d_notes	character	Coffee D - Notes
prefer_abc	character	Between Coffee A, Coffee B, and Coffee C which did you prefer?
prefer_ad	character	Between Coffee A and Coffee D, which did you prefer?
prefer_overall	character	Lastly, what was your favorite overall coffee?
wfh	character	Do you work from home or in person?
total_spend	character	In total, much money do you typically spend on coffee in a month?
why_drink	character	Why do you drink coffee?
why_drink_other	character	Other reason for drinking coffee
taste	character	Do you like the taste of coffee?
know_source	character	Do you know where your coffee comes from?
most_paid	character	What is the most you've ever paid for a cup of coffee?
most_willing	character	What is the most you'd ever be willing to pay for a cup of coffee?
value_cafe	character	Do you feel like you're getting good value for your money when you buy coffee at a cafe?
spent_equipment	character	Approximately how much have you spent on coffee equipment in the past 5 years?
value_equipment	character	Do you feel like you're getting good value for your money when you buy coffee at a cafe?
gender	character	Gender
gender_specify	character	Gender (please specify)
education_level	character	Education Level
ethnicity_race	character	Ethnicity/Race
ethnicity_race_specify	character	Ethnicity/Race (please specify)
employment_status	character	Employment Status
number_children	character	Number of Children
political_affiliation	character	Political Affiliation

Our client ran a survey to better understand the preferences of potential customers for their new line of coffee. The survey includes questions about the potential customers' coffee preferences, demographics, and coffee consumption habits, as well as taste test results of four varieties of coffee.

Our client wants to know if it should recommend a new variety of coffee (**coffee D**) based on customer demographics, preferences, and ratings for a standardized set of three coffe varieties.

Exercise 1

Import the data set and prepare it for modeling. Implement the data preparation we discussed in the last application exercise. Specifically,

- Keep only columns that would plausibly be available without users conducting the taste test and that provide useful, relevant information for modeling.
- Convert categorical variables to an appropriate format.[1]
- Drop observations with missing values for the outcome of interest, `coffee_d_personal_preference`.
- Convert the outcome of interest to a binary variable indicating whether the respondent liked coffee D or not ($\text{Like} > 3$).

Exercise 2

Partition the data set. Split the data into training/test sets. Further partition the training set using an appropriate resampling method. *You will use this resampled set for all model training and evaluation.* Provide a brief explanation for why you chose this specific technique.

Exercise 3

Estimate a null model. Predict a respondent's preferenec for coffee D. Calculate the accuracy, Brier score, and ROC AUC of the model (as well as any other metrics you deem relevant). Interpret the results.

Exercise 4

Evaluate a penalized regression model.

At minimum, the feature engineering recipe should do everything recommended/required for a penalized regression model.

The model should be tuned, at minimum, over the `penalty` and `mixture` tuning parameters.

Beyond that, it is up to you to decide what variables to use as predictors, additional feature engineering steps, tuning parameters, tuning method, etc. You should document these decisions and explain why you made them. Report the results of at least 3 distinct workflows for a penalized regression model. Be sure to include the accuracy, Brier score, and ROC AUCs of the models (as well as any other metrics you deem relevant), and generate a calibration plot for the best performing model.

> **Generating calibration plots**
>
> - R: {probably}
> - Python: sklearn calibration plots

> **Note**
>
> I have established the floor of what you need to do for each exercise. To earn full credit, please think deliberately about your modeling approach and describe/justify it in your submission.

Exercise 5

Evaluate a boosted trees model.

At minimum, the feature engineering recipe should do everything recommended/required for a boosted tree model.

The model should be tuned over an appropriate set of parameters.

Beyond that, it is up to you to decide what variables to use as predictors, additional feature engineering steps, tuning parameters, tuning method, etc. You should document these decisions and explain why you made them. Report the results of at least 2 distinct workflows for a boosted tree model. Be sure to include the accuracy, Brier score, and ROC AUCs of the models (as well as any other metrics you deem relevant), and generate a calibration plot for the best performing model.

Exercise 6

Choose your own adventure. Choose and fit a third model of your choice to predict whether or not the respondent likes coffee D. This can be any model you want, but it must be different from the penalized regression and boosted trees models you already fit. Document your choice of model and justify why you think it is appropriate for this problem. Perform any data preprocessing or feature engineering steps you think are appropriate to improve the model's performance. Tune the model across relevant hyperparameters. How does this model perform?

Exercise 7

Finalize a predictive model. Choose a model from a previous exercise (or try something else) and finalize it. Report the final accuracy, Brier score, and ROC AUC of the model using the test set, and generate a calibration plot for the best performing model. Interpret the performance of your model.

Exercise 8

Your client is excited to hear what you have done for this project and eager to deploy this model ASAP. **What is your recommendation?**

Generative AI (GAI) self-reflection

As stated in the syllabus, include a written reflection for this assignment of how you used GAI tools (e.g. what tools you used, how you used them to assist you with writing code), what skills you believe you acquired, and how you believe you demonstrated mastery of the learning objectives.

> Render, commit, and push one last time.
>
> Make sure that you commit and push all changed documents and your Git pane is completely empty before proceeding.

Wrap up

Submission

- Go to http://www.gradescope.com and click *Log in* in the top right corner.
- Click *School Credentials* → *Cornell University NetID* and log in using your NetID credentials.
- Click on your *INFO 4940/5940* course.
- Click on the assignment, and you'll be prompted to submit it.
- Mark all the pages associated with exercise. All the pages of your homework should be associated with at least one question (i.e., should be "checked").

Grading

- Exercise 1: 4 points
- Exercise 2: 4 points
- Exercise 3: 2 points
- Exercise 4: 10 points
- Exercise 5: 10 points
- Exercise 6: 10 points
- Exercise 7: 5 points
- Exercise 8: 5 points
- GAI self-reflection: 0 points
- Total: 50 points

Footnotes